

Table 1 | Framework for assessing offset quality and climate benefits

Dimension	Indicator	Rule or standard illustratively used, coding framework and data source	Scientific basis in literature
1. Relative quality risks	Do credits come from offset project types with a lower likelihood of overstating their emissions reduction or additionality?	Categorisations of offset project types with a lower, medium and higher risk using the relative quality risks framework in the <i>Quality Offsets Guide</i> by the Stockholm Environmental Institute and GHG Management Institute ¹⁵ .	8,10,12,13,15,16
2. Age	Is the window between the offsetting activity and the time of retirement in line with industry standards?	Rule by Carbon Offsetting and Reduction Scheme for International Aviation (CORSIA) that excludes credits with a vintage year and project start year earlier than 2016.	15,21,32,75
3. Price	Does the credit come from offset project types that typically sell for above-average prices?	Estimates of the average price paid for offsets by project category from Ecosystem Marketplace ⁷⁶ .	11,15,21
4. Country of implementation (applied to renewable energy projects)	Do credits derive from projects implemented in low-income countries where the diffusion of renewable energy is low and hampered by market or policy conditions?	(1) Rule by GS and VCS that limits eligible renewable energy projects to those located in a least developed country (LDC), and (2) Rule by GS that limits eligible renewable energy projects to those located in a low-income country or lower-middle income country where the penetration rate of the proposed energy technology is below 5%. Country classifications and data from World Bank ⁷⁹ and IRENA ⁵⁷ .	12,13,58

Suppliers and buyers of offsets have attempted to circumvent such concerns by claiming to use ‘quality’ offsets^{21,24}. Guidance about offset quality has also emerged from VCM governance frameworks and from best-practice principles proposed by various stakeholders^{25–28}. Despite considerable heterogeneity across different conceptions of quality²⁹, most stress the need to ensure additionality and permanence, avoiding over-crediting and double-counting, and protecting against negative effects on society and the natural environment^{7,15,26,28,30}.

Such quality principles, however, do not guarantee genuine climate benefits. Consider the case of an offset from a wind farm or forest conservation project that started 15 years ago. The climate impact of claiming such an offset today may be compromised in four ways. First, renewable energy projects carry a high risk of overestimating emissions reductions and lacking additionality^{12,13,15}. Second, using historical mitigation activities to offset emissions today fails to promote new decarbonisation activities beyond those already scheduled to occur³¹. Indicators of quality should therefore consider the age of offset projects and credits^{32,33}. Third, quality indicators should also address price. Cheap offsets typically originate from over-credited projects with low additionality¹⁵, diverting funds from projects with higher quality control measures that cost more¹¹. Fourth, in countries where renewable energy has diffused widely and become standard practice, there is a weak argument for additionality. Thus, projects like renewable energy should be implemented in countries where such technologies have not yet mainstreamed due to technological, financial or policy hurdles^{12,13}. Meanwhile, to draw down and permanently sequester historical emissions in pursuit of net-zero emissions, VCM best-practice principles^{34,35} and researchers^{36–38} have emphasised the importance of upscaling investments in carbon removal.

Since offset quality is strongly influenced by characteristics at the individual project level, the extant literature has focused on GHG accounting methodologies and assessing project-level climate benefits^{8,11–13,30,39}. Yet there remains a need for a framework and indicators that can be applied with readily available market-level data to determine if the offsets retired by large corporate buyers correspond with key metrics of high quality and high climate benefits. However, firm-level analyses^{5,40–42} are few, despite many companies depending heavily on offsets to pursue decarbonisation goals. Further, scholars have yet to exhaustively study the publicly available data on registries to examine the behaviour of large-scale corporate offset buyers.

Accordingly, we examine the characteristics and quality of the offsets used by the twenty companies responsible for retiring the

largest volumes on the VCM between January 2020 and December 2023. We build an original dataset that compiles each company’s offset procurements from the VCM’s three largest registries: Verra’s Verified Carbon Standard (VCS), the United Nation’s Clean Development Mechanism (CDM) and Gold Standard (GS). After clarifying the extent to which offset retirements support emissions avoidance or removal, we set out to answer: ‘To what extent could the offsets retired by these companies be considered high quality and likely to benefit the climate?’. We investigate retirement behaviour across four dimensions (Table 1): (1) use of offsets from low/high-risk project types; (2) age of projects and credits; (3) cost of credits and (4) country of implementation (applied only to renewable energy projects).

The analysis reveals that these twenty companies predominantly sourced their offsets from low-quality avoidance activities. Most offsets carry a high risk of overestimating emissions reductions and fail to meet industry standards regarding age and country of implementation. We also find that companies have actively targeted cheap credits. Our results provide further evidence that predominant practices on the VCM are not supporting effective climate mitigation. Our contribution is to demonstrate that individual companies are a major cause of persisting quality issues due to their demand for problematic and cheap offset types known to overstate emission reductions. We also provide the first publicly available dataset compiling the attributes of offsets used by the largest corporate buyers on the VCM. This complements the extant literature’s focus on supply-side issues such as project characteristics and registry methodologies.

Results

Scale and context of offsetting activity

Collectively, the twenty companies examined retired 134 MtCO₂e over 2020–2023 (Fig. 1a). This volume is considerable, representing just over one-fifth of all global retirements made on the three registries (VCS, CDM and GS) during the same period (Table S1).

All companies but one in our sample have used offsets to claim carbon neutrality or pursue decarbonisation targets (Table S2). Domains of offset use extend from operational emissions (e.g. Boeing, Telstra and Norwegian CL) to various claims of ‘carbon-neutral’ products, such as offset-bundled LNG tankers (Shell and Chevron), parcel delivery (DPD and Yamato) and flights (Delta Airlines and easyJet).

The two largest offset buyers are Shell and Delta Airlines, each retiring around 23.5 MtCO₂e over 2020–2023 (Fig. 1a and Table S3a, b). Retirements by these two companies alone make up roughly 35% of all offsets in the data set. Most companies retired far less than this: mean